Loan Default Prediction Using Classification Methods

A credit-risk binary-classification study following the CRISP-DM methodology

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Abstract

In this project, I explored whether loan applicants' information can be used to predict if they are likely to repay their loan or default. I used the Kaggle Credit Risk Dataset, which contains information about 32,581 loan applications. Following the CRISP-DM process, I first explored and prepared the data before training three different classification models: Logistic Regression, a Decision Tree and an Artificial Neural Network. I compared their performance using several measures, including accuracy, precision, recall, F1-score and ROC-AUC. The Decision Tree provided the best overall balance between performance and interpretability, with an accuracy of around 92%. The neural network was better at identifying applicants who might default, achieving the highest recall and ROC-AUC. The analysis showed that loan grade, the loan-to-income ratio and home-ownership status were the most important factors related to default risk.

1. Introduction

Loan default prediction is important because lenders need to decide whether an applicant is likely to repay a loan. In this project, I treat this as a binary classification problem, where the model predicts either that an applicant will repay the loan or that they may default.

My aim is to build and compare several classification models using the applicant's financial, demographic and credit-history information. I am also interested in comparing their accuracy with how easy their decisions are to understand.

I followed the CRISP-DM methodology throughout the project. This included understanding the business problem, exploring and preparing the data, training the models and evaluating their results. Since the dataset contains more non-default cases than default cases, accuracy alone does not give a complete picture of model performance. For this reason, I also used precision, recall, F1-score and ROC-AUC. Recall is especially important here because incorrectly classifying someone who later defaults could be more costly for a lender than incorrectly rejecting an applicant who would have repaid the loan.

2. Dataset

The study uses the publicly available **Credit Risk Dataset** from Kaggle (kaggle.com/datasets/laotse/credit-risk-dataset). It contains **32,581 records** and 12 attributes from demographic, financial to credit-bureau information. The target variable is **loan_status**, where 1 is a default (bad credit risk) and 0 a non-default (good credit risk).

Attribute	Description	Type
person_age	Applicant age (years)	Numeric

person_income	Annual income	Numeric
person_home_ownership	RENT / OWN / MORTGAGE / OTHER	Categorical
person_emp_length	Employment length (years)	Numeric
loan_intent	Purpose of the loan (6 categories)	Categorical
loan_grade	Loan grade A (best) to G (worst)	Ordinal
loan_amnt	Loan amount requested	Numeric
loan_int_rate	Interest rate (%)	Numeric
loan_percent_income	Loan amount as fraction of income	Numeric
cb_person_default_on_file	Historical default on record (Y/N)	Categorical
cb_person_cred_hist_length	Length of credit history (years)	Numeric
loan_status	**TARGET: 1 = default, 0 = non-default**	**Binary**

Table 1. Dataset attributes.

3. Methodology

3.1 Data Understanding & Exploratory Analysis

Before training the models, I explored the dataset to understand its structure and identify any problems that needed to be addressed. I found missing values in loan interest rate, affecting approximately 9.6% of the rows, and in employment length, affecting around 2.7% of the rows. There were also 165 duplicate records. In addition, some values were not realistic, such as ages up to 144 years and employment lengths up to 123 years.

The target variable was imbalanced, with approximately 78% of the applications being non-defaults and 22% being defaults. Because of this imbalance, I decided to use recall, F1-score and ROC-AUC alongside accuracy when evaluating the models.

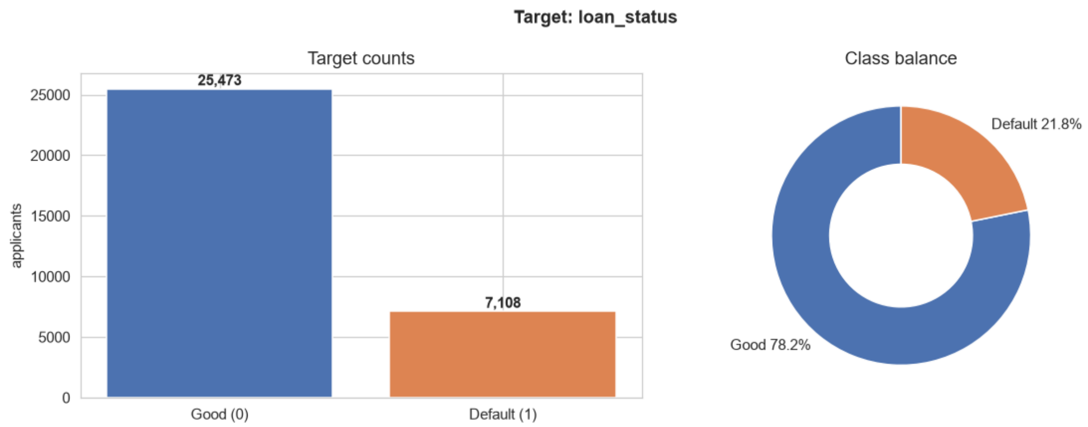

Figure 1. The distribution of the target variable. Most applications are classified as good credit risks, while around 22% are defaults.

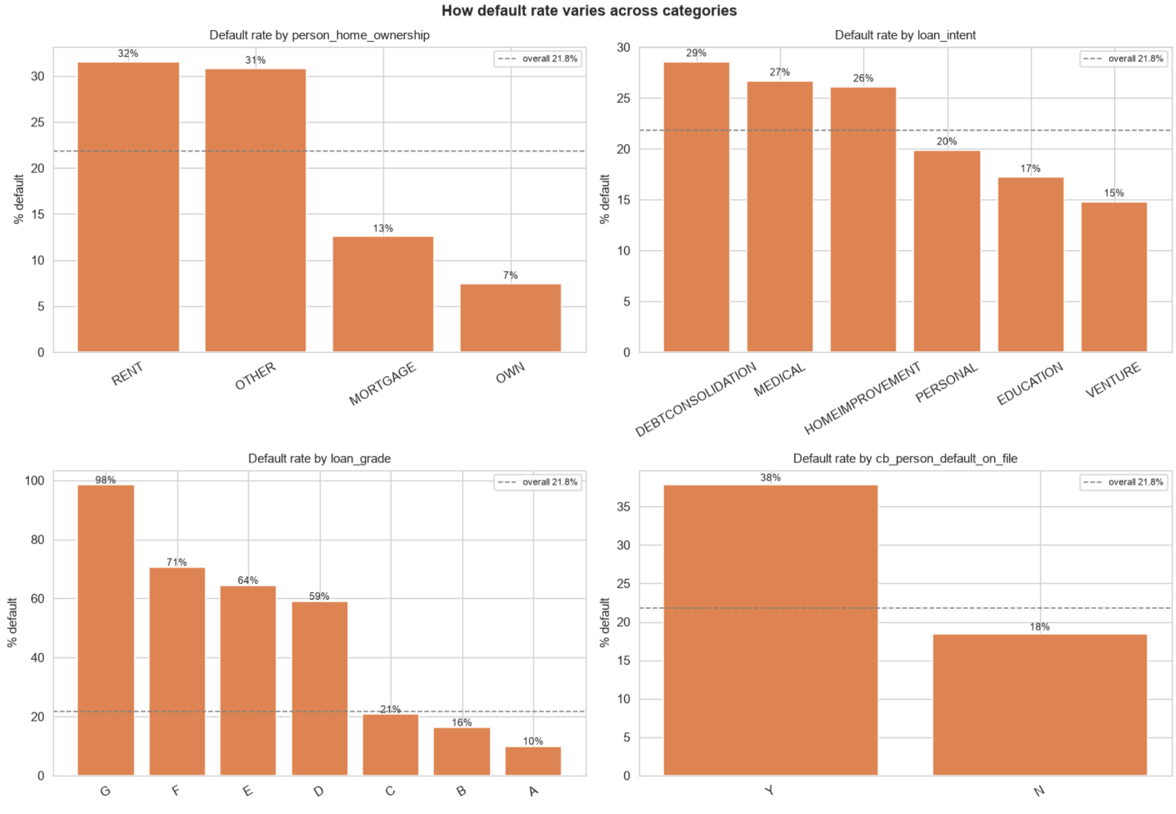

Figure 2. Default rates across the categorical variables. Loan grade appears to be the strongest indicator: the default rate increases from approximately 10% for grade A to 98% for grade G. Renters also have a considerably higher default rate than homeowners.

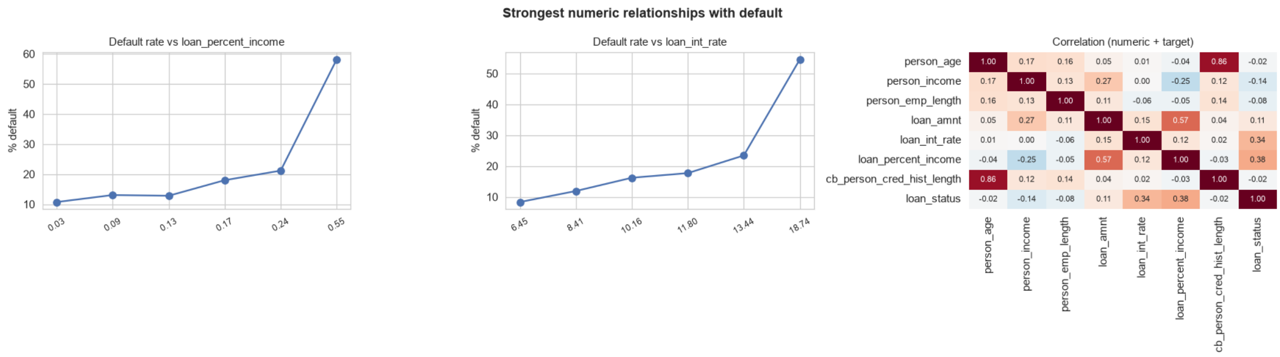

Figure 3. The default rate increases as the loan-to-income ratio and interest rate increase. Among the numeric variables, loan-to-income ratio, with a correlation of 0.38, and interest rate, with a correlation of 0.34, have the strongest relationships with the target.

Overall, the exploratory analysis suggested that default risk does not follow a simple straight-line pattern. Instead, there are noticeable changes across loan grades and loan-to-income bands. This supported my decision to include tree-based and neural-network models alongside Logistic Regression.

3.2 Data Preparation

The following steps were applied, with all imputation and scaling parameters fitted on the training set only to avoid data leakage: (i) removal of 165 duplicate rows and the physically-impossible outliers; (ii) imputation of missing *loan_int_rate* using the median rate per loan grade (grade and rate are tightly linked) and missing *person_emp_length* using the median; (iii) ordinal encoding of *loan_grade* (A→G as 0→6), binary encoding of the Y/N default flag, and one-hot encoding of *person_home_ownership* and *loan_intent*; (iv) a stratified 80/20 train/test split; and (v) standardisation of features for the models that require it. The Logistic Regression and neural network use standardised inputs, while the Decision Tree uses the raw (scale-invariant) values.

3.3 Modelling

I trained three different models. I used the same training and test data for each model so that their results could be compared fairly.

First, I trained a Logistic Regression model. I used this as a baseline because it is a simple model and its results are relatively easy to understand. It also helped me see how much better the other models performed.

The second model was a Decision Tree with a maximum depth of six. I chose this model because it can show its decisions through a series of rules. This makes it easier to understand why the model classifies an applicant as more or less likely to default.

Finally, I trained an Artificial Neural Network with two hidden layers. The layers contained 32 and 16 units. I included this model because it can identify more complicated patterns in the data. I also used early stopping so that the model would not continue training if its performance on new data stopped improving.

4. Results and Evaluation

The results for all three models are shown in Table 2. Overall, the Decision Tree and the neural network performed much better than Logistic Regression. Logistic Regression achieved an accuracy of about 85%, but its recall was only 0.50. This means that it identified only about half of the applicants who actually defaulted.

The Decision Tree achieved an accuracy of 91.96%, while the neural network achieved 92.01%. The Decision Tree had the highest precision, meaning that most of the applicants it classified as potential defaulters really were defaults. The neural network performed better in terms of recall, F1-score and ROC-AUC, although the difference between the two models was quite small.

Model	Accuracy	Precision	Recall	F1	ROC-AUC
Logistic Regression	0.849	0.725	0.501	0.593	0.864
Decision Tree	0.920	**0.972**	0.652	0.780	0.905
ANN (MLP)	**0.920**	0.918	**0.697**	**0.792**	**0.916**

Table 2. Performance of the three models on the test set. The best result for each metric is highlighted.

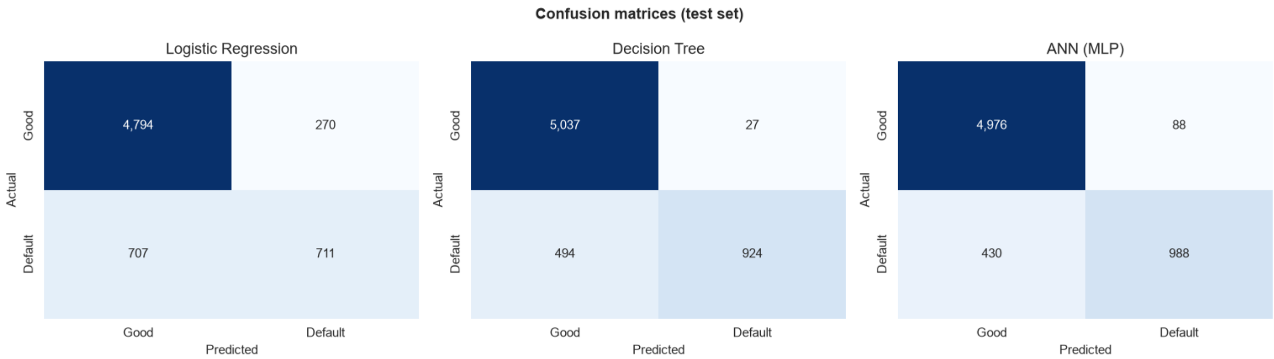

Figure 4. Confusion matrices for the three models. Logistic Regression missed more defaults than the Decision Tree and the neural network.

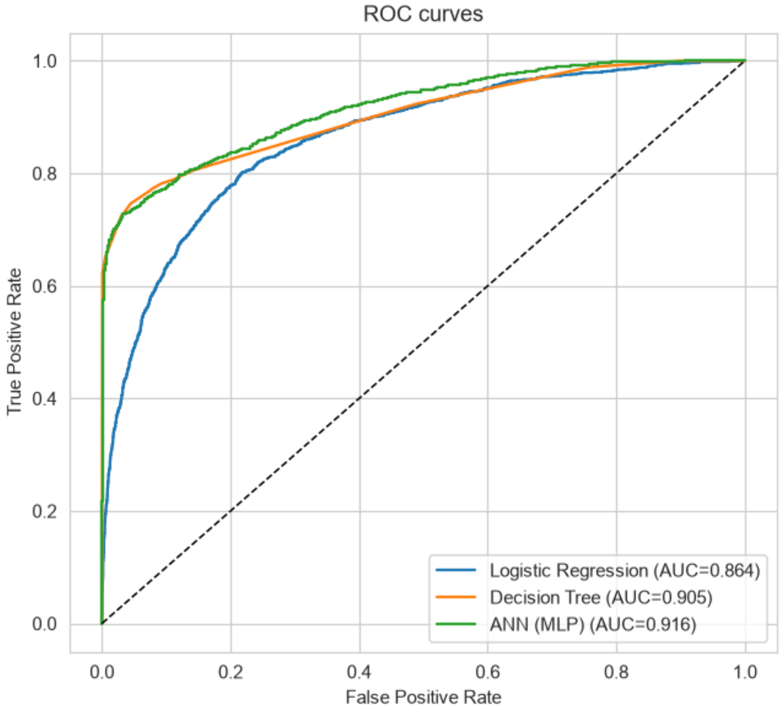

Figure 5. ROC curves for the three models. The neural network achieved the highest ROC-AUC, followed by the Decision Tree.

The Decision Tree also made it possible to look at the rules behind its predictions. The most important features were the loan-to-income ratio, loan grade and whether the applicant rented their home. This was similar to what I found during the exploratory analysis. In general, applicants with a high loan-to-income ratio who rented were more likely to be classified as defaulters. Applicants with a lower loan-to-income ratio, a better loan grade and sufficient income were more likely to be predicted as non-defaults.

5. Discussion

When I compared the three models, the main thing I noticed was that Logistic Regression was not able to capture all the patterns in the data. It was the simplest model to understand, but it missed quite a lot of the applicants who actually defaulted. This suggests that the connection between the applicant information and loan default is not completely linear.

The Decision Tree worked much better and was still relatively easy to follow. I could look at its rules and get an idea of why an applicant was classified as a likely defaulter. This makes the model easier to explain than the neural network.

The neural network gave the strongest results for recall, F1-score and ROC-AUC, but it was much harder to see how it reached its decisions. Personally, I think the Decision Tree is a good compromise for this project. Its results were close to those of the neural network, but its predictions were easier to understand. However, if the main priority was to find as many possible defaulters as possible, the neural network would probably be the better option.

6. Conclusion and Future Work

In this project, I used the CRISP-DM process to explore the Kaggle Credit Risk Dataset and compare three different models for predicting loan default. The results showed that the Decision Tree and the neural network performed better than Logistic Regression. This suggests that loan default is influenced by several factors working together, rather than by one simple pattern.

The most important factors appeared to be loan grade, the loan-to-income ratio and home-ownership status. Based on the results, I think the Decision Tree is the most suitable model for this project because it performed well while still being possible to understand and explain. The neural network would be useful if identifying as many potential defaults as possible was the main goal.

There are several ways this project could be improved in the future. I could try methods such as class weighting or SMOTE to deal with the imbalance in the target variable. I could also tune the models more carefully using cross-validation and compare them with methods such as Random Forest or gradient boosting. Another useful improvement would be to adjust the decision threshold depending on whether missing a potential defaulter or incorrectly rejecting a good applicant is considered more costly.

Online Resources and References

[1] Project source code. GitHub. https://github.com/EvgeniiaEliseeva/Machine_learning_project

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